Shivan Poudel

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PROFESSIONAL SUMMARY:

Computer Engineering at Rose-Hulman, with minor in Artificial Intelligence. Experience in experimental and theoretical physics, including research at CERN and independent software projects. Currently learning the foundations of Machine Learning and Neural Networks; experience building AI applications and analysis of scientific datasets. Looking to contribute to the development of Artificial Intelligence and quantitative fields.

EXPERIENCE:

Research Assistant Dec 2024 – Feb 2026

CERN – ATLAS Higgs Group

- Processed and analyzed approximately 1,000 real and simulated Large Hadron Collider (LHC) collision events specializing in Higgs to diphoton decay channels using Python and ROOT.
- Contributed to research within the ATLAS Collaboration, whose work was recognized by the 2025 Breakthrough Prize in Fundamental Physics.

Research Assistant Oct 2026 – Present

Rose-Hulman Institute of Technology

- Investigate the electronic structure of carbon nanotube samples through carbon K-edge X-ray absorption near-edge structure (XANES) spectroscopy, analyzing spectral features and variations across experimental conditions.
- Conduct comparative spectral analysis using Python and Matplotlib, identifying potential X-ray beam shifts and investigating inconsistencies across experimental measurements.

PROJECTS:

AI Image Recognition with NVIDIA Technology

- Built a PyTorch-based AI application using ResNet-18 to classify car images by brand and model with ~97% accuracy using open datasets and deep learning techniques.
- Leveraged NVIDIA GPU acceleration for model training, applying computer vision methods to optimize classification performance.

Translation App

- Built a Python-based macOS menu bar app for instant translation using global hotkeys and automatic language detection.
- Integrated clipboard management and system-level event handling into a standalone desktop application; currently expanding features and language support.

Research Paper on Noether's Theorem May 2025 – Oct 2025

- Authored a theoretical physics paper on Noether's Theorem, proving conservation laws (momentum, charge, energy) stem from local symmetries.
- Applied tensor calculus, Lagrangian mechanics, scalar fields, and quantum field theory throughout the analysis.

EDUCATION:

Bachelor of Science, Computer Engineering (Minor, Artificial Intelligence) May 2030

Rose-Hulman Institute of Technology – Terre Haute, IN

Current Coursework: Calculus III, Physics II

High School Diploma May 2026

Dougherty Valley High School – San Ramon, CA

Relevant Coursework: AP Physics C: Mechanics, AP Calculus AB, AP Calculus BC, AP Computer Science A, AP Chemistry, AP Statistics,

SKILLS:

Programming & Tools: Python, PyTorch, Java, ROOT, LaTeX

Technical Skills: Computer Vision, Data Analysis, Scientific Writing, Research Collaboration, Calculus, Physics

ACTIVITIES & LEADERSHIP:

Virtual Tutoring Program June 2025 – Mar 2026

Tutor

- Tutored students in Nepal; designed lessons introducing advanced mechanics, electricity and magnetism, and calculus to bridge academic gaps.

NorCal Volleyball Club June 2021 – July 2024

- Competed at the national level, winning medals at AAU tournaments and multiple regional events; served as team captain for 2 years, leading strategy and team development.